MATHS MANTHAN

• TUITION CLASSES •

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Subject: Mathematics | **CBSE CLASS 10 – PERIODIC ASSESSMENT TEST – 08** | **Time:** 60 Minutes

Class: Class X (CBSE) | **Chapter 4: Quadratic Equations** | **MM: 30**

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CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, correct algebraic substitution, zero skipped steps, and final boxed answers with proper units.
2. In alternative (**OR**) sub-questions, complete step-by-step solutions and equal step marking apply to both options.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 2M]**. Roots rejected by physical domain/restrictions are explicitly stated.

Official Evaluation Key | Maths Manthan Meerut

Section A: MCQ & Assertion-Reason Questions

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Q1. If the quadratic equation $x^2 - kx + 9 = 0$ has equal roots, then the value of k is: [1]

(A) ± 3 **(B)** ± 6

(C) 6 only **(D)** 9

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (B) ± 6

Comparing $x^2 - kx + 9 = 0$ with $ax^2 + bx + c = 0$, we have $a = 1$, $b = -k$, and $c = 9$.

Formula Used: Equal real roots $\iff D = b^2 - 4ac = 0$

$$(-k)^2 - 4(1)(9) = 0 \implies k^2 = 36 \implies \boxed{k = \pm 6}$$

Hence, option **(B)** is correct. **[1 mark]**

Q2. The number of real roots of the quadratic equation $(x-1)^2 - 4 = 0$ is: [1]

(A) 1 **(B)** 2

(C) 0 **(D)** 3

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (B) 2

Solving $(x-1)^2 - 4 = 0$:

Formula Used: $A^2 - B^2 = (A - B)(A + B)$

$$(x-1)^2 = 4 \implies x - 1 = \pm 2 \implies x = 3 \text{ or } x = -1$$

There are $\boxed{2}$ distinct real roots (3 and -1). Hence, option **(B)** is correct. **[1 mark]**

Q3. A quadratic equation whose roots are $(3+\sqrt{2})$ and $(3-\sqrt{2})$ is: [1]

(A) $x^2-6x+7=0$ **(B)** $x^2+6x+7=0$

(C) $x^2-6x-7=0$ **(D)** $x^2+3x-2=0$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: **(A)** $x^2-6x+7=0$

Given roots $\alpha=3+\sqrt{2}$ and $\beta=3-\sqrt{2}$:

Formula Used: $x^2-(\alpha+\beta)x+\alpha\beta=0$

$$\alpha+\beta=(3+\sqrt{2})+(3-\sqrt{2})=6, \qquad \alpha\beta=(3+\sqrt{2})(3-\sqrt{2})=3^2-(\sqrt{2})^2=9-2=7$$

Therefore, the equation is $\boxed{x^2-6x+7=0}$. Hence, option **(A)** is correct. **[1 mark]**

Q4. If $ax^2+bx+c=0$ has equal roots, then the value of each equal root is: [1]

(A) $-\frac{b}{a}$ **(B)** $-\frac{b}{2a}$

(C) $\frac{b}{2a}$ **(D)** $\frac{c}{a}$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: **(B)** $-\frac{b}{2a}$

By the quadratic formula:

Formula Used: $x=\frac{-b\pm\sqrt{D}}{2a}$

For equal roots, the discriminant $D=b^2-4ac=0$. Substituting $D=0$:

$$x=\frac{-b\pm\sqrt{0}}{2a}=\boxed{-\frac{b}{2a}}$$

Hence, option **(B)** is correct. **[1 mark]**

Q5. Which of the following quadratic equations has no real roots? [1]

(A) $x^2-4x+3\sqrt{2}=0$ **(B)** $x^2+4x-3\sqrt{2}=0$

(C) $x^2-4x-3\sqrt{2}=0$ **(D)** $3x^2+4\sqrt{3}x+4=0$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: **(A)** $x^2-4x+3\sqrt{2}=0$

A quadratic equation has no real roots if and only if $D=b^2-4ac<0$.

Formula Used: No real roots $\iff D=b^2-4ac<0$

Checking each option:

(A) $D=(-4)^2-4(1)(3\sqrt{2})=16-12\sqrt{2}\approx 16-16.97=-0.97<0$ (no real roots).

(B) $D=16+12\sqrt{2}>0$ (two distinct real roots).

(C) $D=16+12\sqrt{2}>0$ (two distinct real roots).

(D) $D=(4\sqrt{3})^2-4(3)(4)=48-48=0$ (two equal real roots).

Hence, option **(A)** is correct. **[1 mark]**

Q6. Assertion-Reason: [1]

Directions: *(A) Both A & R true, R explains A; (B) Both true, R does NOT explain A; (C) A true, R false; (D) A false, R true.*

Assertion (A): If 2 is a root of the equation $x^2 + bx + 12 = 0$, then $b = -8$.

Reason (R): A real number α is a root of the quadratic equation $p(x) = 0$ if and only if $p(\alpha) = 0$.

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A) Both Assertion (A) and Reason (R) are true, and Reason (R) is the correct explanation of Assertion (A).

Applying Reason (R) to $p(x) = x^2 + bx + 12$ with root $\alpha = 2$:

Formula Used: $p(2) = 0 \implies (2)^2 + b(2) + 12 = 0$

$$4 + 2b + 12 = 0 \implies 2b = -16 \implies b = -8$$

Thus Assertion (A) is **true**, Reason (R) is **true**, and Reason (R) correctly explains Assertion (A). **[1 mark]**

Section B: Very Short Answer Questions

MATHS MANTHAN | $2 \times 2 = 4$ Marks

Q7. Find the value of k for which the equation $(k-4)x^2 + 2(k-4)x + 2 = 0$ has equal roots. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

For $(k-4)x^2 + 2(k-4)x + 2 = 0$ to be quadratic, $k - 4 \neq 0 \implies k \neq 4$.

Here $a = k - 4$, $b = 2(k-4)$, and $c = 2$.

Formula Used: Equal real roots $\iff D = b^2 - 4ac = 0$

Setting $D = 0$:

$$[2(k-4)]^2 - 4(k-4)(2) = 0$$
$$4(k-4)^2 - 8(k-4) = 0 \quad \textbf{[1 mark]}$$
$$4(k-4)[(k-4) - 2] = 0 \implies 4(k-4)(k-6) = 0 \implies k = 4 \text{ or } k = 6$$

We reject $k = 4$ because the x^2 term vanishes giving $2 = 0$ (impossible).

Hence, $\boxed{k = 6}$. **[1 mark]**

Q8. Solve the following quadratic equation for x by factorization: $2x^2 + x - 6 = 0$. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Given equation: $2x^2 + x - 6 = 0$. Here $a = 2$, $b = 1$, $c = -6$, so $ac = -12$.

Formula Used: Split $+x = +4x - 3x$ since $4 \times (-3) = -12$ and $4 - 3 = 1$

$$2x^2 + 4x - 3x - 6 = 0$$
$$2x(x+2) - 3(x+2) = 0 \quad \textbf{[1 mark]}$$
$$(x+2)(2x-3) = 0$$

Equating each factor to zero:

$$x + 2 = 0 \implies x = -2 \qquad \text{or} \qquad 2x - 3 = 0 \implies x = \frac{3}{2}$$

Hence, the roots are $\boxed{x = \frac{3}{2} \text{ and } x = -2}$. **[1 mark]**

Q9. Solve for x: $3\left(\frac{3x-1}{2x+3}\right) - 2\left(\frac{2x+3}{3x-1}\right) = 5, \quad x \neq -\frac{3}{2}, \frac{1}{3}$. **[3]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Let $y = \frac{3x-1}{2x+3}$ (well-defined and non-zero since $x \neq -\frac{3}{2}, \frac{1}{3}$). Then $\frac{2x+3}{3x-1} = \frac{1}{y}$.

Formula Used: Substitute $y = \frac{3x-1}{2x+3} \implies 3y - \frac{2}{y} = 5$

Multiplying throughout by y:

$$3y^2 - 2 = 5y \implies 3y^2 - 5y - 2 = 0 \quad \textbf{[1 mark]}$$

Factorising $(3 \times (-2) = -6;\ -6 + 1 = -5)$:

$$3y^2 - 6y + y - 2 = 0 \implies 3y(y-2) + 1(y-2) = 0$$
$$\implies (y-2)(3y+1) = 0 \implies y = 2 \text{ or } y = -\frac{1}{3} \quad \textbf{[1 mark]}$$

Now back-substituting $y = \frac{3x-1}{2x+3}$:

- **Case 1** $(y = 2)$**:** $\frac{3x-1}{2x+3} = 2 \implies 3x - 1 = 4x + 6 \implies x = -7.$
- **Case 2** $(y = -\frac{1}{3})$**:** $\frac{3x-1}{2x+3} = -\frac{1}{3} \implies 9x - 3 = -2x - 3 \implies 11x = 0 \implies x = 0.$

Both values satisfy $x \neq -\frac{3}{2}, \frac{1}{3}$. Hence, $\boxed{x = -7 \text{ and } x = 0}$. **[1 mark]**

Q10. Seven years ago, Varun's age was five times the square of Swati's age. Three years hence, Swati's age will be two-fifths of Varun's age. Determine their present ages. **[3]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Let Swati's age 7 years ago be x years $(x > 0)$.
Then Varun's age 7 years ago was $5x^2$ years.

Formula Used: Present ages: Swati $= (x + 7)$ yrs, Varun $= (5x^2 + 7)$ yrs

Three years hence from now (10 years after 7 years ago):

- Swati's age $= (x + 7) + 3 = (x + 10)$ years.
- Varun's age $= (5x^2 + 7) + 3 = (5x^2 + 10)$ years. **[1 mark]**

Given that 3 years hence, Swati's age will be $\frac{2}{5}$ of Varun's age:

$$x + 10 = \frac{2}{5}(5x^2 + 10) = 2x^2 + 4$$
$$2x^2 - x - 6 = 0 \quad \textbf{[1 mark]}$$

Factorising $(2 \times (-6) = -12;\ -4 + 3 = -1)$:

$$2x^2 - 4x + 3x - 6 = 0 \implies 2x(x-2) + 3(x-2) = 0 \implies (x-2)(2x+3) = 0 \implies x = 2 \text{ or } x = -\frac{3}{2}$$

Since age cannot be negative, we reject $x = -\frac{3}{2}$. Thus $x = 2$.
Present age of Swati $= 2 + 7 = 9$ years, and present age of Varun $= 5(2)^2 + 7 = 27$ years. (Check: $9 + 3 = 12 = \frac{2}{5}(27 + 3)$.)
Hence, $\boxed{\text{Swati's present age} = 9 \text{ years, Varun's present age} = 27 \text{ years}}$. **[1 mark]**

Q11. A motorboat whose speed is 15 km/h in still water goes 30 km downstream and returns back to the starting dock in a total time of 4 hours 30 minutes. Determine the speed of the river stream. [5]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 5 Marks]

Let the speed of the river stream be x km/h ($0 < x < 15$).
Speed downstream $= (15 + x)$ km/h; Speed upstream $= (15 - x)$ km/h.
Distance $= 30$ km, Total round-trip time $= 4$ h 30 min $= 4\frac{1}{2}$ h $= \frac{9}{2}$ h. **[1 mark]**

Formula Used: Downstream speed $= 15 + x$, Upstream speed $= 15 - x$, $t_{\text{down}} + t_{\text{up}} = \frac{9}{2}$

Setting up the equation:

$$\frac{30}{15+x} + \frac{30}{15-x} = \frac{9}{2}$$
$$30\left[\frac{(15-x)+(15+x)}{(15+x)(15-x)}\right] = \frac{9}{2} \quad \textbf{[1 mark]}$$
$$\frac{30 \times 30}{225 - x^2} = \frac{9}{2} \implies \frac{900}{225-x^2} = \frac{9}{2} \implies \frac{100}{225-x^2} = \frac{1}{2} \quad \textbf{[1}\tfrac{1}{2}\textbf{ marks]}$$

Cross-multiplying:

$$225 - x^2 = 200 \implies x^2 = 25 \implies x = \pm 5 \quad \textbf{[1 mark]}$$

Since speed of the stream cannot be negative ($x > 0$), we reject $x = -5$. (Check: $\frac{30}{20} + \frac{30}{10} = 1.5 + 3 = 4.5$ h.)
Hence, the speed of the river stream is $\boxed{5 \text{ km/h}}$. [$\frac{1}{2}$ **mark**]

Q12. An aircraft departed 50 minutes later than its scheduled departure time. In order to reach its destination 1250 km away on schedule, the pilot had to increase the speed by 250 km/h from its usual speed. Determine the scheduled speed of the aircraft and the original flight duration. [5]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 5 Marks]

Let the usual (scheduled) speed of the aircraft be v km/h ($v > 0$).
Increased speed $= (v + 250)$ km/h, distance $= 1250$ km, and delay $= 50$ min $= \frac{5}{6}$ h. **[1 mark]**

Formula Used: Time $= \frac{\text{Distance}}{\text{Speed}}$, $t_{\text{scheduled}} - t_{\text{actual}} = \frac{5}{6}$ hour

Setting up the time equation:

$$\frac{1250}{v} - \frac{1250}{v+250} = \frac{5}{6}$$
$$1250\left[\frac{(v+250)-v}{v(v+250)}\right] = \frac{5}{6} \implies \frac{1250 \times 250}{v^2 + 250v} = \frac{5}{6} \quad \textbf{[1}\tfrac{1}{2}\textbf{ marks]}$$

Dividing both sides by 5:

$$\frac{250 \times 250}{v^2 + 250v} = \frac{1}{6} \implies v^2 + 250v = 62500 \times 6 = 375000 \implies v^2 + 250v - 375000 = 0 \quad \textbf{[1 mark]}$$

Splitting the middle term ($750 \times (-500) = -375000$ and $750 - 500 = 250$):

$$v^2 + 750v - 500v - 375000 = 0 \implies (v + 750)(v - 500) = 0 \implies v = -750 \text{ or } v = 500 \quad \textbf{[1 mark]}$$

Since speed cannot be negative ($v > 0$), we reject $v = -750$. Thus scheduled speed $v = 500$ km/h.
Original flight duration $= \frac{1250}{500} = 2.5$ hours $= 2$ hours 30 minutes.
Hence, $\boxed{\text{Scheduled speed} = 500 \text{ km/h, Original duration} = 2 \text{ hours } 30 \text{ minutes}}$. [$\frac{1}{2}$ **mark**]

Q13. Case Study: Municipal Swimming Pool & Dual Water Inflow Dynamics

The municipal sports authority is commissioning an Olympic-standard training pool. Two automated intake pipes are installed to fill the empty reservoir. Working together, both pipes fill the pool in $9\frac{3}{8}$ hours. The pipe of larger bore diameter takes 10 hours less than the pipe of smaller bore diameter to fill the pool when operating independently. [4]

(i) If the smaller pipe takes x hours alone, write an expression for the portion of the pool filled by both pipes together in 1 hour. **[1]**

(ii) Formulate the quadratic equation in x in standard form modeling the filling process. **[1]**

(iii) Find the time required by each pipe separately to fill the pool. **OR** Determine what fraction of the pool remains empty if both pipes operate simultaneously for only 5 hours. **[2]**

▶ CASE-BASED SOLUTION & MARKING SCHEME: [Total: 4 Marks]

Formula Used: Portion filled in 1 hour $= \dfrac{1}{\text{Time taken to fill alone}}$

(i) Combined 1-Hour Fill Expression: Smaller pipe takes x hours ($x > 10$), so it fills $\dfrac{1}{x}$ in 1 h. Larger pipe takes $(x-10)$ hours, so it fills $\dfrac{1}{x-10}$ in 1 h. Thus, portion filled by **both pipes together in 1 hour** is:

$$\boxed{\frac{1}{x} + \frac{1}{x-10} = \frac{2x-10}{x(x-10)}} \quad \left(\text{which also equals } \frac{8}{75} \text{ per hour}\right) \quad \textbf{[1 mark]}$$

(ii) Quadratic Equation in Standard Form: Together they fill the pool in $9\frac{3}{8} = \frac{75}{8}$ hours, so combined 1-hour work $= \dfrac{8}{75}$:

$$\frac{2x-10}{x^2-10x} = \frac{8}{75}$$

$$75(2x-10) = 8(x^2-10x) \implies 150x - 750 = 8x^2 - 80x$$

$$8x^2 - 230x + 750 = 0 \implies \boxed{4x^2 - 115x + 375 = 0} \quad \text{(dividing by 2)} \quad \textbf{[1 mark]}$$

(iii) First Part (Individual Filling Times): Factorising ($4 \times 375 = 1500$; $-100 - 15 = -115$):

$$4x^2 - 100x - 15x + 375 = 0 \implies 4x(x-25) - 15(x-25) = 0$$

$$\implies (x-25)(4x-15) = 0 \implies x = 25 \text{ or } x = \frac{15}{4} = 3.75$$

If $x = 3.75$ h, the larger pipe would take $3.75 - 10 = -6.25$ h < 0 (impossible), so $x = \frac{15}{4}$ is rejected.
Thus smaller pipe time $= 25$ hours and larger pipe time $= 25 - 10 = 15$ hours.
Hence, $\boxed{\text{Smaller pipe} = 25 \text{ hours, Larger pipe} = 15 \text{ hours}}$. **[2 marks]**

OR (Alternative Part — Empty Fraction After 5 Hours): Combined 1-hour rate $= \dfrac{8}{75}$ of pool.

Fraction filled in 5 hours $= 5 \times \dfrac{8}{75} = \dfrac{8}{15}$, so fraction remaining empty $= 1 - \dfrac{8}{15} = \boxed{\dfrac{7}{15}}$. **[2 marks]**

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